Bal Bharati Hackathon 3.0

Idea Submission

NIVRA: Before Impact

Spot warning signs. Help staff respond sooner.

Category: Cat2

Problem Statement Title	CCTV can miss warning signs before an incident escalates
Theme	T4 | Community & Civic Innovation
Team Name	Team Damage Control
Team Leader	Harsh Jain
Unit Name	Bal Bharati Public School, Pitampura
Team Size	5 students (excluding mentor)

Your Idea

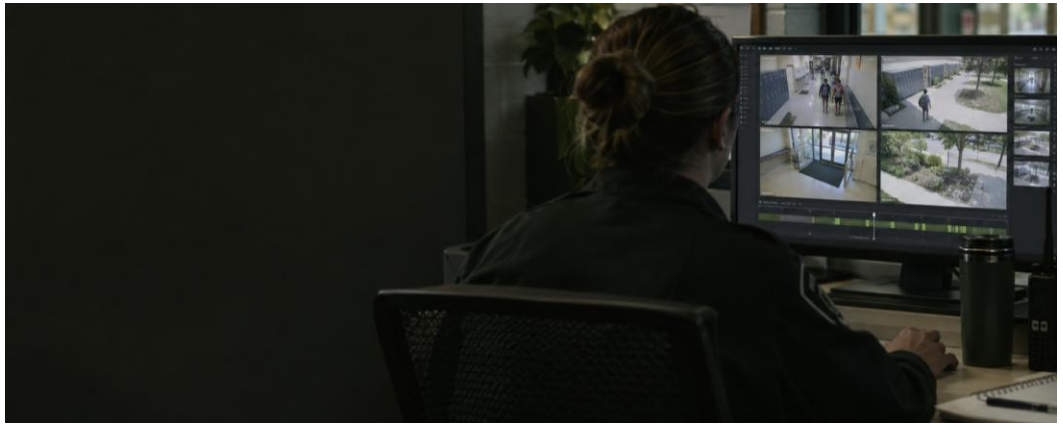
THE PROBLEM



A camera can record a fight.
Who notices it starting?

- Staff cannot watch every feed at once.
- A developing incident can go unnoticed.

NIVRA'S RESPONSE



NIVRA draws attention
to moments worth checking.

- It flags concerning movement.
- A member of staff reviews the alert.

Visible Warning Signals

WHAT TRIGGERS A REVIEW?



Movement needs context.

- Repeated, directed arm movements
- The distance between people
- Several signals over time

A PERSON MAKES THE CALL

01 / Notice the alert

NIVRA highlights a pattern worth checking.

02 / Look at the scene

Staff assess what is actually happening.

03 / Decide what to do

Dismiss it, keep watching, or arrange help.

Technical Approach

Inputs

Camera, local video or scripted scenario.

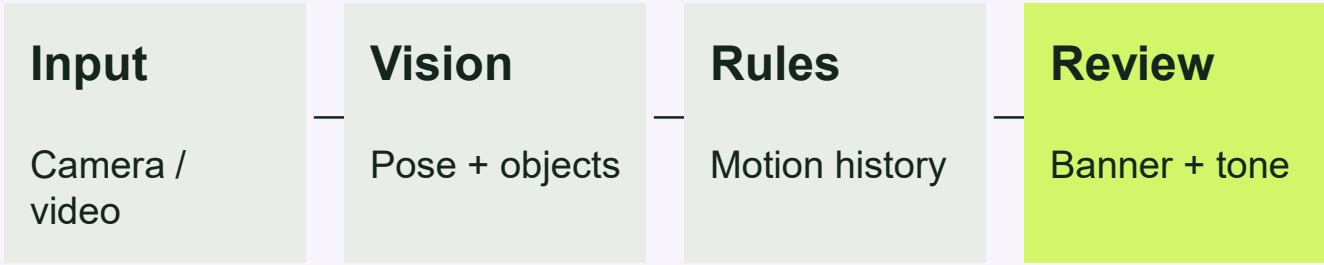
Two Web Workers

One for pose, one for objects. Local MediaPipe + WASM.

Runtime

GPU for real-time pose when supported. CPU for review scans and fallback.

BROWSER ARCHITECTURE



Frames remain in the browser

- Video is processed on the device.
- No footage is uploaded or recorded by the app.
- Temporary tracks clear when the session ends.

Technical Approach | Demo

Try the current lab

1. Choose camera, video or test scenario.
2. Inspect skeletons and review reasons.
3. Acknowledge alerts; pause, replay or stop.

211 tests passed

Lint and production build also passed during local verification.

CURRENT LAB AND NEXT STEPS



Proposed next: connected response

- Connect cameras across a site
- Replay the previous 8 seconds
- Help staff contact an available responder

Feasibility and Viability

FEASIBILITY OF THE IDEA

- Runs in a browser on a laptop or phone.
- Uses camera input or a local video file.
- Start with one supervised school trial before connecting more cameras.

POTENTIAL CHALLENGES & RISKS

01 Sports and play can trigger reviews.

02 Occlusion or weak devices reduce coverage.

03 School video needs permission and privacy controls.

STRATEGIES TO OVERCOME THEM

01 Require human review; test positive and quiet clips.

02 Use Lite / CPU fallback; show visibility and runtime state.

03 Keep frames local; obtain school approval before trials.

Impact and Benefits

Social Impact	Economic Impact	Environmental Impact
Help staff notice a developing incident and decide when to step in.	Build on existing cameras where possible, reducing the need for new equipment.	Reuse suitable devices and extend their working life.

BENEFITS OF THE SOLUTION

- For staff: fewer feeds to search when an alert appears.
- For schools: a practical way to trial earlier incident review.
- Next step: measure missed events, false alerts and review time.

Impact and Benefits | Why it matters

4.4 million	1.25 million	~12,000 / day
Global injury-related deaths in 2019, including unintentional injuries.	Violence-related injury deaths in 2019, including self-harm and conflict.	Approximate daily equivalent of 4.4 million. Not a count of public fights.

WHAT THESE FIGURES MEAN FOR NIVRA

- Our focus is narrower: help staff notice a developing incident sooner.
- Test whether alerts are useful, timely and easy to review.
- These global figures describe the problem; they do not prove NIVRA works.

Research and References

EVIDENCE AND SOURCES

NIVRA repository
Implementation, test fixtures and limitations.

WHO | Injuries and violence
Broad global burden; not product impact.

MediaPipe Pose Landmarker
Landmark API and model documentation.

MoE school safety / DPDP Act
Policy context and privacy review before trials.

Axis Object Analytics
Existing event-analytics category for comparison.

Technology Stack

IMPLEMENTED IN THE REPOSITORY

React 19 + TypeScript + Vite
Browser UI and production build.

MediaPipe Tasks Vision 1.0.1
Pose Landmarker Full / Lite, float16.

EfficientDet-Lite0
Optional object analysis on CPU.

Web Workers + WASM + Canvas
Two workers; GPU attempt / CPU fallback.

Vitest + Playwright + Oxlint
Unit tests, browser tests and lint tools.

Team Member Details

Team leader: Harsh Jain

Student's Details

SL	Student ID	Name	Class	DOB	Email	Phone Number	Gender
1	223564	Harsh Jain	12C	23/04/2009	223564@pp.balbharati.org	9810141505	Male
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Mentor Details

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